

STRUTHIO ONE SLIM

Build manual · Windows 11 · every step in build order



16.5 mm thin. One click to flash, one soldering job, one screw size. The handheld checks itself the first time you switch it on.

Rev S2 · commit 2f17878 · 2026-10-02

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How to read this manual

Do the parts in order. Each step has a picture, what to do, and a green ✓ **check**: don't go on until it's true. Orange boxes are the things that matter most.

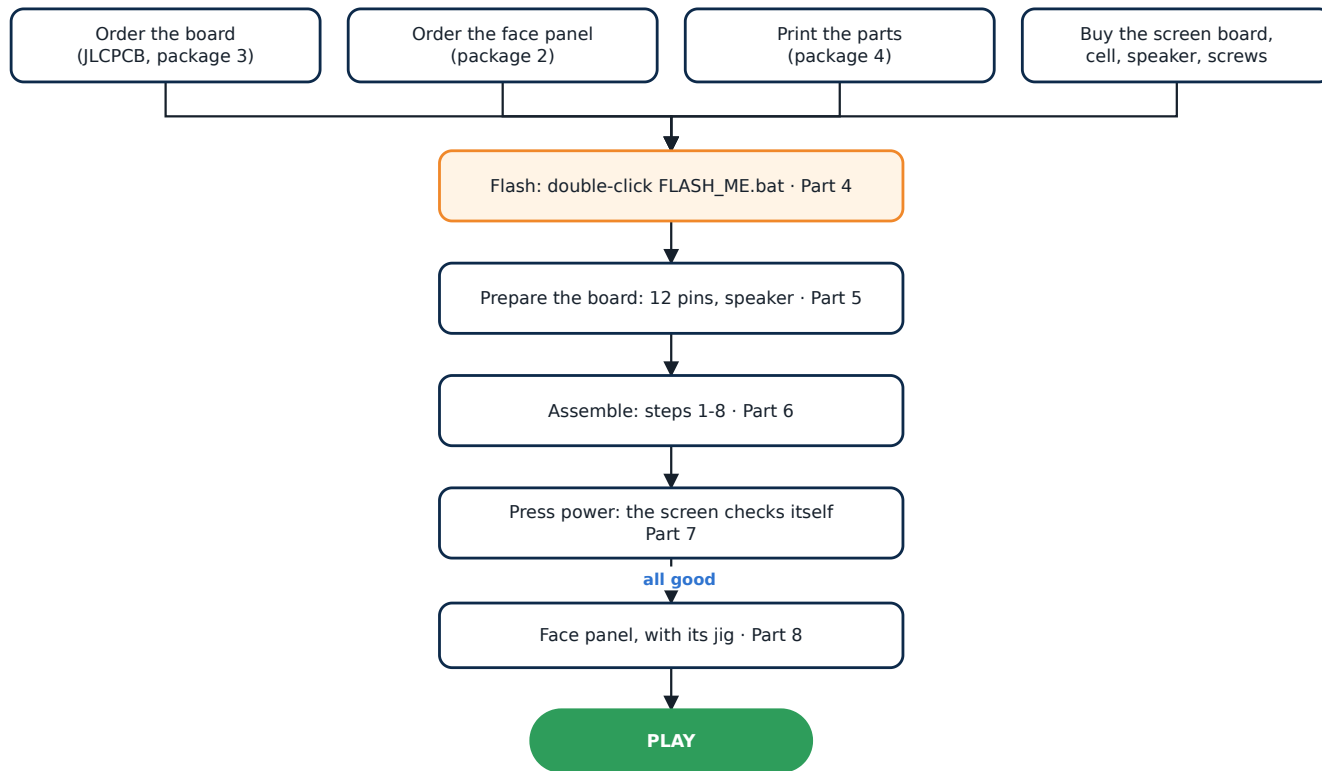
Pictures are labelled: **SCREENSHOT** real output of the real firmware and tools, **FROM THE 3D MODEL** rendered from the exact case and board design (the parts each step adds glow orange), **DIAGRAM** drawn from the design's own numbers.

The packages are referred to by number: **1** firmware, **2** face panel, **3** board (JLCPCB), **4** 3D print, **5** documents (this manual). The shopping list on its own: **STRUTHIO_ORDER** in package 5.

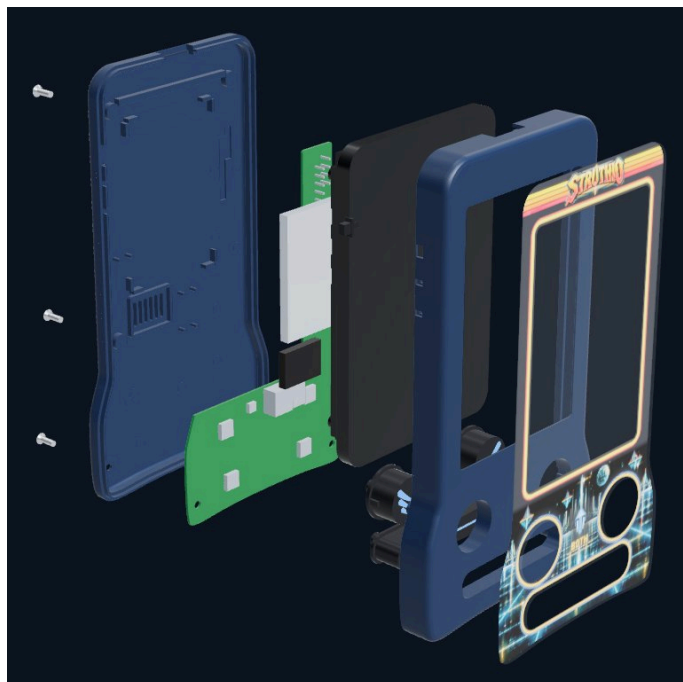
The whole build at a glance

PART 1

Four things are made or bought at the same time. When the screen board arrives you put the game on it, solder 12 pins and a speaker, build, switch on: the handheld checks itself.



The build, start to finish DIAGRAM



Everything that goes into it, front to back. The animated version is `slim_assembly.gif` in package 5 FROM THE 3D MODEL

What you need

PART 2

Buy

✓	Qty	Item	Exactly what
☐	1	Screen board	Waveshare ESP32-S3-Touch-LCD-3.5B , the bare board (not "-C"). Its speaker lead comes with it.
☐	1	Battery	LiPo 302535 , about 250 mAh, with protection board, JST PH 2.0 mm 2-pin plug.
☐	1	Speaker	Same Sky (CUI) CMS-151125-078SP , 15 × 11 × 2.5 mm (Mouser, DigiKey).
☐	5	Screws	M2 × 6 countersunk (flat head). All five the same.
☐	1	Header strip	2.54 mm male header, single row (you use 12 pins).
☐	1	Foam tape	Double-sided, 1.0 mm thick, about 20 × 30 mm.
☐	1	USB-C data cable	One that carries data (a phone's own cable usually does).

Made for you (Part 3)

✓	Item	From
☐	ONE SLIM board	JLCPCB, assembled, 0.8 mm (package 3)
☐	Face panel	an acrylic shop (package 2)
☐	8 printed parts	your printer or a print service (package 4)

Tools

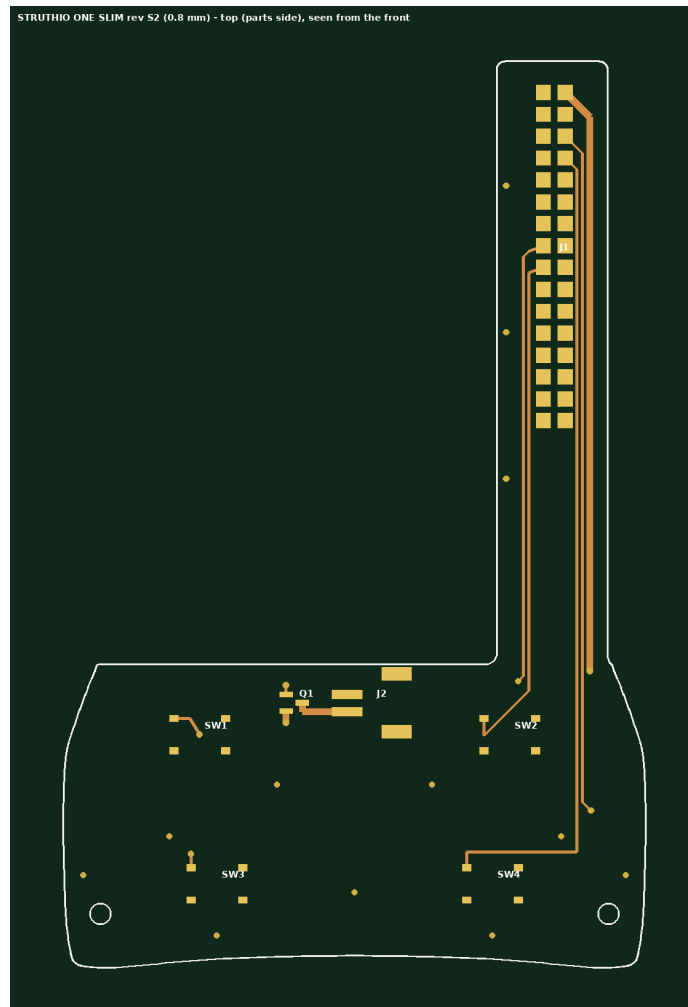
- ☐ A soldering iron and solder (12 pins and 2 pads)
- ☐ Flush cutters
- ☐ A small Phillips screwdriver (PH0)
- ☐ A paperclip (only for re-flashing through the case later)
- ☐ A Windows 11 computer with internet access

Order and make the parts

PART 3

The ONE SLIM board (package 3, folder ONE_SLIM)

1. On jlcpcb.com upload `struthio_one_slim_gerbbers.zip`. Set **PCB Thickness: 0.8 mm**.
2. Turn on **PCB Assembly**, top side. Upload `BOM_JLCPCB.csv` and `CPL_JLCPCB.csv` : six small parts, all surface-mount.
3. In the placement preview, **J2**'s opening faces the board's **right** edge (compare with the picture). Rotate in 90° steps if not.



The board as ordered (top side, seen from the front of the handheld) DIAGRAM

The face panel (package 2)

Send an acrylic shop the files and this sentence: **1.0 mm clear cast acrylic, cut to the DXF, reverse-printed (colour, then white) from the mirrored PNG, window left clear; clear adhesive transfer tape (3M 468MP) laminated on the back, except over the window.**

The printed parts (package 4, folder ONE_SLIM)

Front shell, back shell, two wing buttons, rocker, power button, pin jig, panel jig. PETG (or ASA), 0.2 mm layers, 4 walls, 6 top and bottom layers, 40 % infill. No printer? Upload the STLs to a print service and ask for **MJF nylon (PA12)** or **PETG**; not resin, which is too brittle for the screws.

Side views, the bed at the bottom. No supports needed.



How the shells and buttons sit on the bed [DIAGRAM](#)

Put the game on the board

PART 4 · AS SOON AS THE SCREEN BOARD ARRIVES, BEFORE YOU BUILD IT IN

4.1 Unzip the firmware package

Right-click `STRUTHIO_ONE_1_Firmware_Flashing.zip` → **Extract All...** → `C:\` → Extract. You get `C:\struthio`.

Not inside OneDrive, and no spaces in the path.

4.2 Plug the board in

Connect the WaveShare board to the computer with the USB-C **data** cable. Nothing else is connected to it yet.

4.3 Double-click FLASH_ME.bat

In `C:\struthio`, double-click **FLASH_ME.bat**. It installs Python and the flashing tool the first time (it may take a few minutes and ask you to allow it), finds the board by itself, writes the game, its pictures and its music, and checks every byte.

If Windows says "Windows protected your PC": click **More info**, then **Run anyway**. It is a plain text file you can read in Notepad.

✓ The window ends with **DONE**.

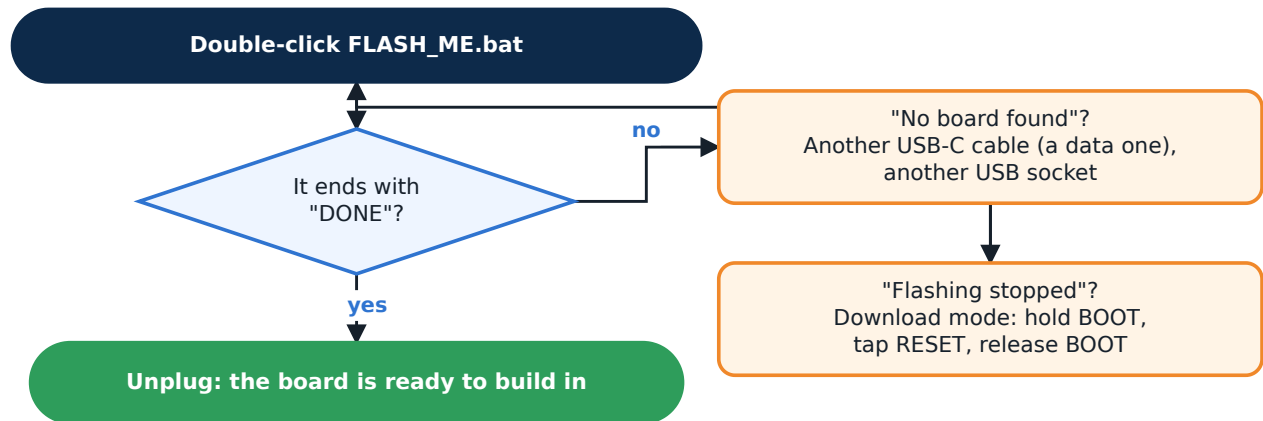
```
Terminal ~/struthio (the emulator bench)

maker@bench:~/struthio$ ./flash_me.sh
STRUTHIO - put the game on the board
[1/4] Python: ok
[2/4] esptool: ok
[3/4] Board found on socket://localhost:5555
[4/4] Writing the game, its pictures and its music (about 2 minutes)...
...
Connected to ESP32-S3 on socket://localhost:5555:
...
Writing 'struthio.bin' at 0x00010000...
...
Hash of data verified.
Writing '../handheld/build/assets/struthio.pak' at 0x00410000...
...
Hash of data verified.
Writing '../handheld/build/assets/struthio_music.ima' at 0x00d10000...
...
Hash of data verified.
...
DONE. Every file was written and checked. Unplug the board, or press RESET: the first power-on ch...
```

The real output of the same flasher (`flash_me.sh`, `FLASH_ME.bat`'s twin for macOS and Linux, with the same steps and messages), run against Espressif's ESP32-S3 emulator. **1** the board found; **2** each part written and checked; **3** done.

Lines left out are marked ... SCREENSHOT

If it doesn't work



Flashing problems DIAGRAM

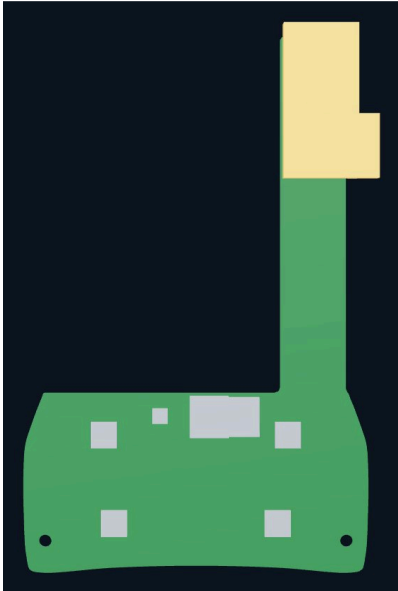
Download mode: hold the board's **BOOT** key, press and release **RESET**, release BOOT. FLASH_ME asks for it when it needs it. In the finished case the two keys are behind the pin holes on the left side (Part 9).

Prepare the board

PART 5 · THE ONE SOLDERING JOB: 12 PINS AND 2 SPEAKER PADS

5.1 Cut two header strips

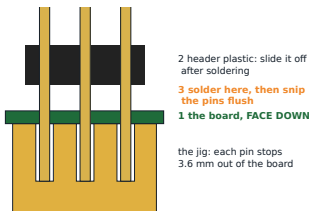
From the header strip, cut a strip of **4** pins and a strip of **8** pins (snip between pins).



5.2 Board face down in the jig

Lay the ONE SLIM **face down** (buttons down) on the printed **pin jig**, its strip inside the jig's fence. Two small pegs go into two holes: it only sits flat the right way round.

✓ The board lies flat in the jig; the back silkscreen reads "J1: PINS 1-7 ODD + 4-18 EVEN".



Side view of the jig DIAGRAM

5.3 Pins in, solder, plastic off, snip

Push the 4-pin strip into the outer row's top four holes and the 8-pin strip into the inner row (the jig only has holes where pins go), **long side first**, until they stop. Solder each pin on the back. Slide the black plastic off the pins with pliers, then snip every pin flush with the board.

✓ 12 pins stand 3.6 mm out of the front, straight; nothing sticks out at the back.

5.4 The speaker

Cut the Waveshare speaker's lead off its old speaker, leaving it long. Strip and tin the two wires and solder them to the CMS-151125's two pads, either way round (the amplifier drives both).

✓ Two clean joints; the lead's plug still on its other end.

Assemble

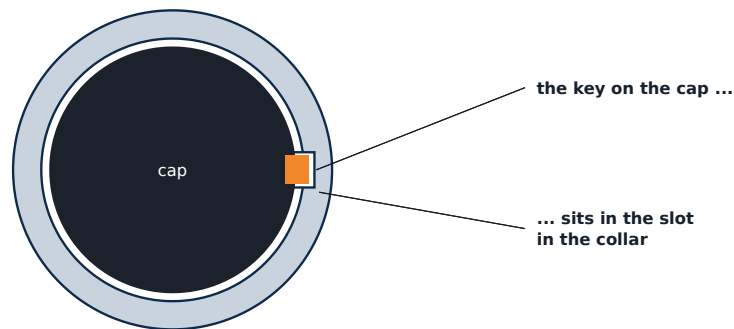
PART 6 · STEPS 1-8 · ON A SOFT CLOTH, NOTHING NEEDS FORCE



1 Wing buttons

From the **inside** of the front shell, push each round cap into its collar, its key in the slot on the collar's outer side.

✓ Both caps in, wings upright from the front, each springs back.



A wing button seen from inside: the key points away from the middle, so it only goes in one way DIAGRAM



2 Rocker

Push the rocker into its collar from the inside. Cut its axle: push 1.75 mm filament into the pin jig's slot to the closed end and cut it at the open end (exactly 11 mm). Push the axle through the hole in the collar's top wall until it stops.

✓ The rocker tips both ways and springs back.



3 Power button

From the inside, push the small power button through the slot in the **left** wall (left as you hold the handheld), its wide flange inside.

- ✓ The button stands 0.8 mm out of the side and slides freely.



4 Lay in the screen board

Screen **down**, **USB-C at the top**. The glass drops into its pocket in the front shell.

- ✓ It lies flat; the power button's tip touches its PWR key; press the button: you feel a click.



5 The ONE SLIM

Pins down, lower the board straight onto the screen board: the pins go into the long black socket (J8) on its right side, and the board rests on the shell's posts.

- Line the pins up first, then press evenly. If they won't go in, look: one pin may be bent.

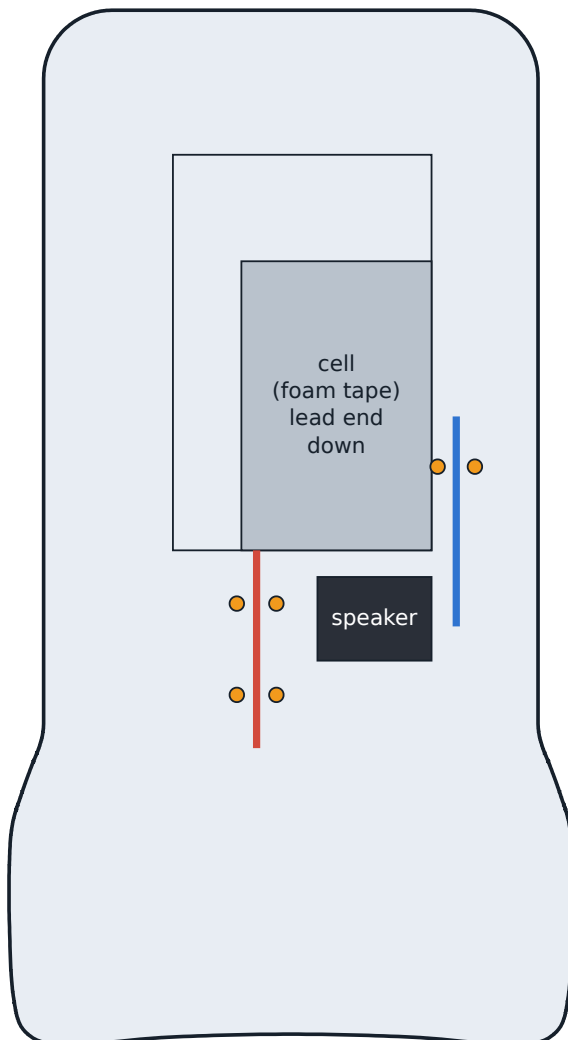
- ✓ The board sits flat on its posts.



6 Back shell: speaker and cell

Press the speaker into its lip in the back shell, face toward the grille. Stick the foam tape on the cell's back and press the cell into the bay's corner (see the diagram), lead end down. Press each lead under its clips.

✓ Speaker in its lip, cell stuck down, leads under their clips.



Seen from inside the back shell,
USB-C end at the top.

red: the cell lead, under two clips,
down to J2 on the board

blue: the speaker lead, under a clip,
up to J9 on the Waveshare

gold dots: the lead clips

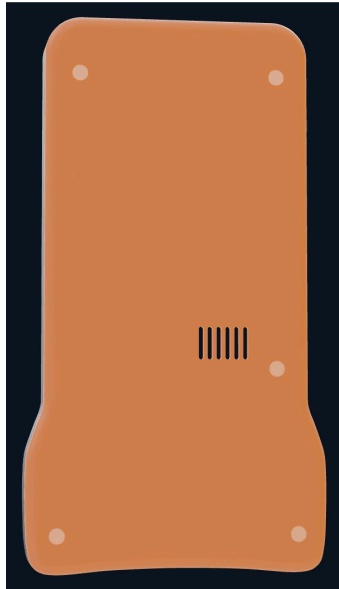
Where things go in the back shell **DIAGRAM**

7 Plug in

Speaker plug into **J9** on the screen board (small 2-pin socket at its left edge). Battery plug into **J2** on the ONE SLIM (red wire to +). The plugs only go in one way.

The battery is now connected: the handheld can switch on if you press the power button.

✓ Both plugs fully in.



8 Close the case

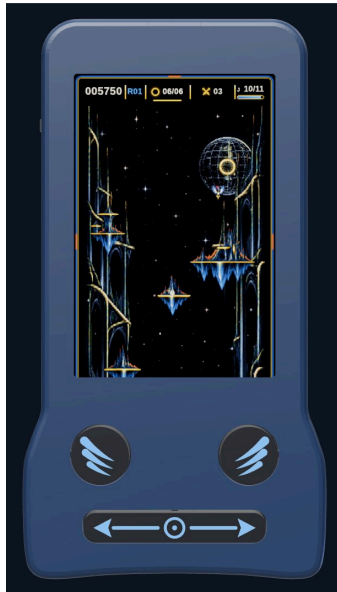
Lay the back shell on: its tongue goes into the front shell's groove all round. Fit the **5 × M2 × 6** screws: three go into the screen board's own metal standoffs, two into the bottom corners. Snug, not tight.

If it doesn't close by itself, open it and look: a lead is in the way. Never force it.

✓ Closed all round, screw heads flush.

Switch on: the first power-on check

PART 7



9 Press the power button

Press the button on the left side. The very first time, hold it until the screen lights (up to 2 s); after that a short press is enough.

✓ The screen lights with STRUTHIO · FIRST POWER-ON CHECK.



10 Follow the screen

Press each wing button, each end of the rocker and the power button **once**: each turns **OK** and clicks (the click is the speaker working). BATTERY shows the cell. Picture upside down? Hold LEFT 2 s. When everything is OK, press **both wings**: the game starts.

STUCK DOWN next to a button means its cap rubs: Part 10. NO BATTERY: check the J2 plug.

✓ ALL GOOD, then the game.

The check, all done (example values)

SCREENSHOT



Power from now on **DIAGRAM**

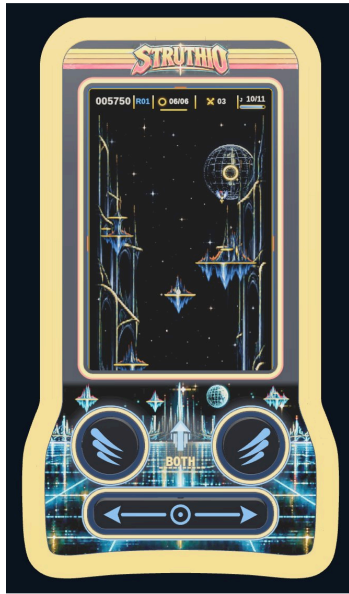
11 Charge

Plug USB-C in, on or off. The cell charges at 100 mA: about 3 hours from empty.

- ✓ Plugged in, it keeps running and never switches itself off.

Fit the face panel

PART 8 · LAST



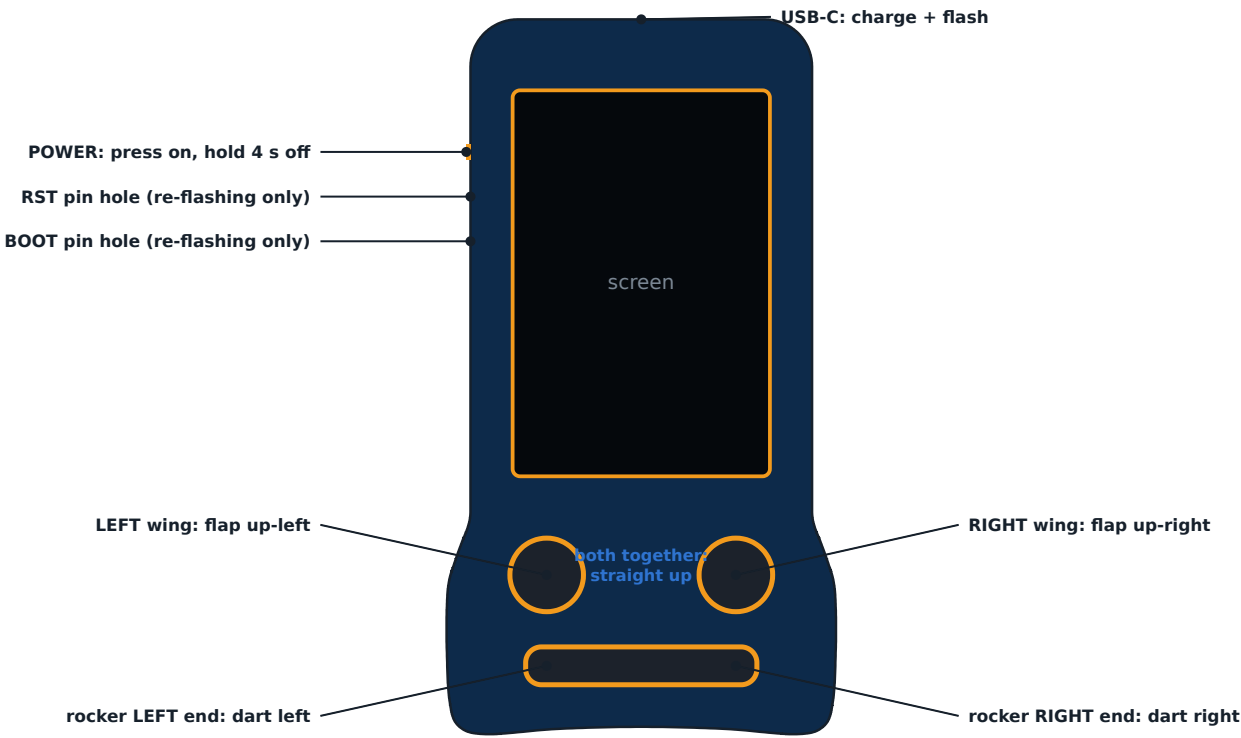
12 Panel, with its jig

Drop the printed panel jig over the front of the case (its skirt hugs the sides). Peel the panel's back liner, drop the panel into the jig's window, press it down from the middle outward, lift the jig off. Peel the front film.

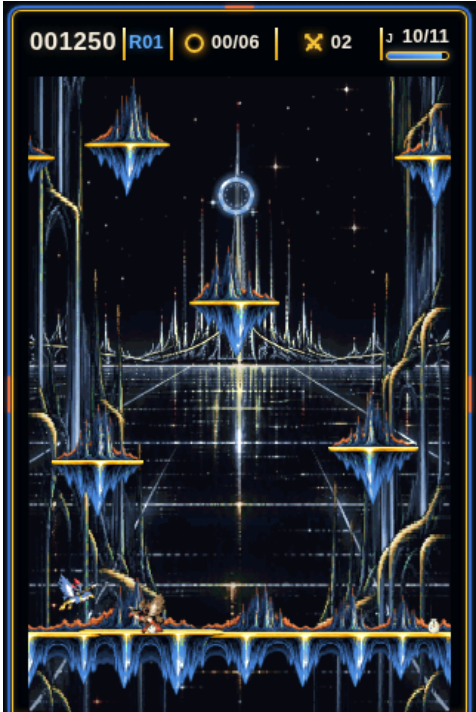
- ✓ The panel sits flat; every button moves freely in its well.

How to play

PART 9



The controls, and what is on the sides DIAGRAM



The game running SCREENSHOT

The controls

Press	Does
LEFT wing	flap up and to the left
RIGHT wing	flap up and to the right
Both together	flap straight up
Hold a wing	steer
Rocker left / right end	aim-assisted dart, that way
Both after GAME OVER	new run
Power button	press: on; hold 4 s: off
Both held while switching on	service mode

Updating the game later

Plug USB-C in and double-click FLASH_ME.bat again. If it asks for download mode: a paperclip in the **BOOT** hole, tap **RST** with another, release BOOT.

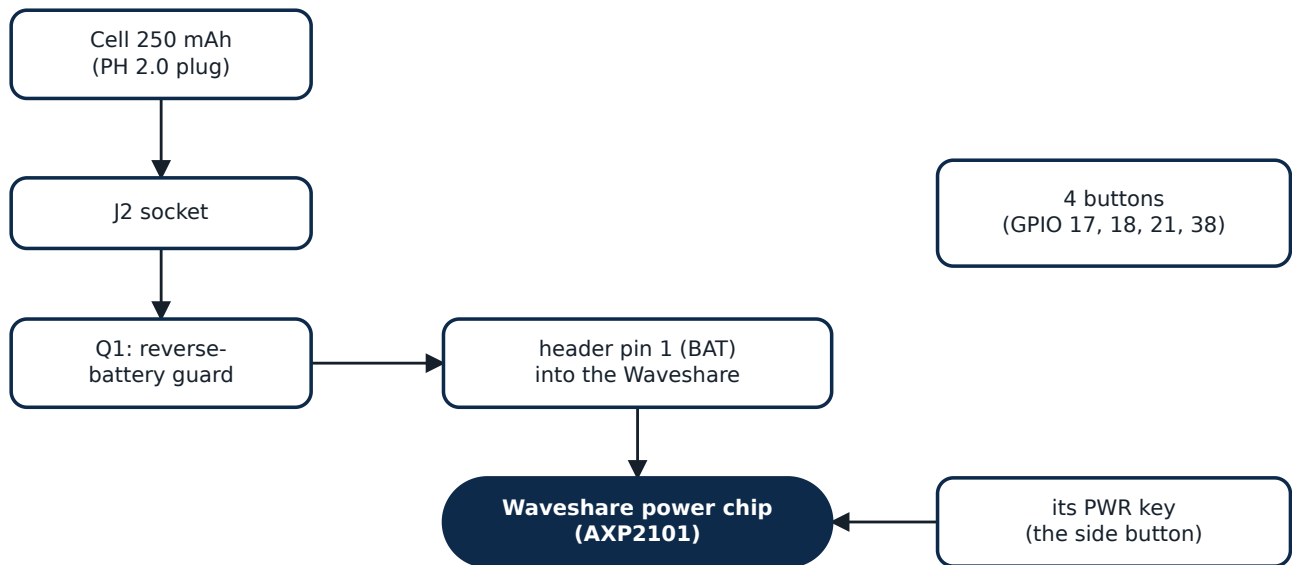
If something is wrong

PART 10

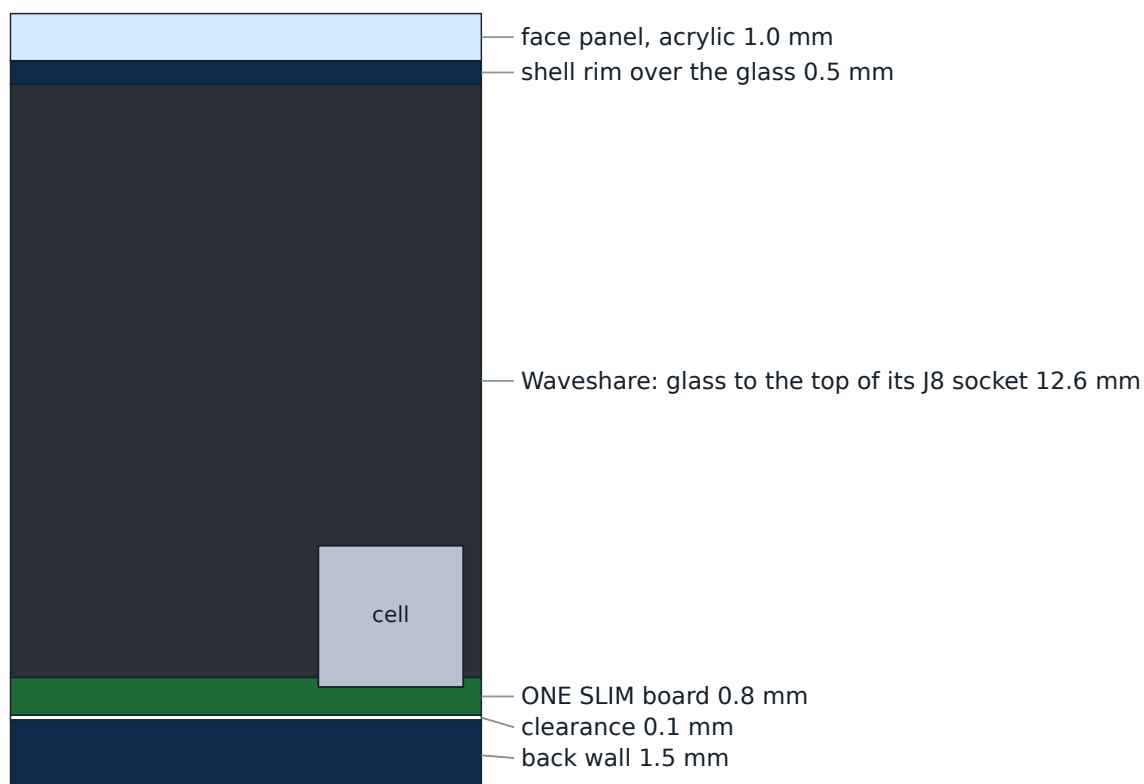
What you see	What to do
Nothing when you press power	Hold it 2 s (the very first time). Battery plug in J2, red to +? Plug USB-C in: if it starts on USB, let the cell charge.
STUCK DOWN on the check screen	That cap's stem touches its switch: take the cap out, sand 0.2 mm off the stem tip.
A button never turns OK	Is the ONE SLIM flat on its posts, every pin in J8? Does the cap move freely?
POWER never turns OK	Does the power button click its key? Push it in from outside: it must move freely.
No click, no sound	Speaker plug in J9? Speaker face against the grille? Speaker pads soldered?
NO BATTERY	J2 plug fully in? Red to +.
Back shell won't close	A lead is in the way: under its clips? Never force it.
Picture upside down	On the check screen: hold LEFT 2 s. Later: service mode, hold RIGHT 1 s.
Flashing fails	Part 4, "If it doesn't work".

How it works

PART 11



The electrics [DIAGRAM](#)



total 16.5 mm

The 16.5 mm stack, front to back (the cell sits behind the screen board's low parts, beside the ONE SLIM) [DIAGRAM](#)

Header pins

1 BAT · 3, 4 GND · 5 GPIO21 (rocker L) · 7 GPIO38 (rocker R) · 16 GPIO17 (left wing) · 18 GPIO18 (right wing). Pin 9 stays empty: that is how the firmware knows it is a SLIM (100 mA charge, power button).

The other pins of the even strip land on empty pads.

Things the first builds should confirm

What	Designed for	If different
J8 socket height	12.6 or 11.5 mm behind the glass: both work	pins go 3.6 or 2.5 mm in
Standoffs threaded M2	the 3D model says SMT M2 nuts	if a screw won't bite, use a slightly longer one
Power button reaches its key	tip 0.2 mm from it, 0.6 mm travel	a drop of glue + 0.3 mm shim on the tip
302535 size	3.0 × 25 × 38 mm with its board	any 3.0 mm cell up to 34 × 52 mm fits the bay

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